Applications of the Normal Distribution Worksheet

1. The medicine entry exam marks are normally distributed with mean 70 and standard deviation 10. Only students with marks higher than 90 are qualified to study medicine. What is the percentage of students that qualifies? If 2000 students take the exam, how many of them will not qualify?
2. The height of an 18-year-old male Australian is known to be normally distributed with mean 170 and standard deviation 16 (in cm). Pilots are required to be between 170 and 190 cm tall. If 2000 18-year-olds apply to become a pilot, approximately how many of them meet the height requirement?
3. Find an estimate for a such that $P(Z < a) = 0.95$ using interpolation.
   The reaction time of a person is found to be normally distributed with mean 400 and standard deviation 150 (in milliseconds). What would be the reaction time needed to be among the top 5% fastest-reacting people?
4. The number of days it rains per year is normally distributed with mean 40 and standard deviation 25. Find the probability that:
   a. There will be at least 60 rainy days next year?
   b. There will be no more than 30 rainy days next year?
   c. There will be between 25 and 45 rainy days next year?
5. The distance a car can travel with a full tank of fuel without refuelling is normally distributed with mean 400 km and standard deviation 125km. A car is considered efficient if it can travel over 700km without refuelling. What percentage of cars are efficient?
   The government pays car owners $1000 per year if they drive efficient cars. There are 10 million cars in Australia. How much would the government pay to those car owners every year?
6. The new Tesla Cybertruck can travel 600km without having to recharge. What percentage of the cars does the Cybertruck beat?
7. Phone prices are getting ridiculous nowadays. It is found that they are normally distributed with mean $1000 and standard deviation $400. A new iPhone 11 Pro Max costs $1800. About what percentage of phones on the market is more expensive than the new iPhone?
8. The number of goals a football striker scores per season is normally distributed with standard deviation 5. A striker than scores more than 9 goals per season is in the top 21.19%. What is the mean number of goals?
   There are 5000 professional strikers in the world. If Harry scores 20 goals per season, approximately how many strikers are better than him?
9. An hour after drinking a beer, a person's blood alcohol content is normally distributed with mean 0.03 and standard deviation 0.01. It is illegal to drive with a blood alcohol content

0.05 or higher. If 100 people leave the company Christmas party an hour after drinking, approximately how many of them are not legally allowed to drive?

10. Sergio is playing a casino game which costs \$10 per try. He can win a random amount that is normally distributed with mean \$5 and standard deviation \$4. What is the probability that he makes profit?
11. The highest temperature on a hot summer day is normally distributed with mean 35°C and standard deviation 2°C. If summer is 90 days in total, how many days would be cooler than 32°C?
12. If the percentage of days over 30°C is higher this year than last year, the government will finally recognise climate change and take action. If last year's temperature had mean 28°C and standard deviation 2.5°C and this year's mean is 29°C with standard deviation 2°C, will the government take action against climate change?
13. A company has a machine that fills orange soda bottles. It dispenses soda according to a normal distribution with mean 600 mL and standard deviation 5 mL. Regulation requires companies to fill their bottles above the advertised volume at least 95% of the time. What volume should be labelled on the bottles? (refer to Q3)
The company also produces sparkling water in bottles labelled 1200 mL. If the machine has the same 5 mL standard deviation, what should the mean value be set at so that the product meets the regulation?
14. A learner driver is required to complete 120 hours of driving to be eligible for the driving test. It is known that the actual hours learner drivers spend on driving are normally distributed with mean 100. Therefore, only about 30.85% of learners actually meet the requirement. What is the standard deviation of hours driven?
15. If X and Y are both normally distributed with mean μ_x, μ_y and standard deviations σ_x, σ_y respectively, then the random variable $X + Y$ is also normally distributed with mean $\mu_x + \mu_y$ and standard deviation $\sqrt{\sigma_x^2 + \sigma_y^2}$.
Roger's exam results are normally distributed. His 3U results have mean 80 and standard deviation 3. His 4U results have mean 70 and standard deviation 4. Roger will receive the high achiever award if the combined marks of his 3U and 4U results is larger than 160. What is the probability that he will receive the award?

Solutions:

Q1: 2.28%, 1954, Q2: 789, Q3: 1.65, 153 milliseconds, Q4 a: 0.2119, b: 0.3446, c: 0.305, Q5: 0.82%, \$82 million, Q6: 94.52%, Q7: 2.28%, Q8: 5, 7, Q9: 2, Q10: 11.51%, Q11: 6, Q12: yes, Q13: 592 mL, 1208.25 mL, Q14: 40, Q15: 2.28%